

UA / UC Detection Rules

Safety Intelligence Framework — Problem Statement 26165

1. Unsafe Act (UA)

Definition: An Unsafe Act is a person's action or behavior that violates a safety procedure, ignores a required control, or increases exposure to a hazard.

Rule ID	Unsafe Act Rule	Example
UA-01	PPE not used	Grinding without required eye protection
UA-02	Safety procedure not followed	Entering confined space without following entry procedure
UA-03	LOTO / isolation not followed	Maintenance started without proper isolation
UA-04	Safety barrier bypassed	Operator bypassed an interlock
UA-05	Unauthorized operation	Untrained person operated equipment
UA-06	Unsafe positioning	Worker stood under a suspended load
UA-07	Exclusion zone violated	Worker entered a restricted lifting area
UA-08	Fall protection not used	Working at height without required harness
UA-09	Unsafe equipment operation	Equipment operated outside safe procedure
UA-10	Safety warning ignored	Operation continued despite an alarm/warning

2. Unsafe Condition (UC)

Definition: An Unsafe Condition is a physical, environmental, equipment, or system condition that creates a hazard or increases the possibility of an incident.

Rule ID	Unsafe Condition Rule	Example
UC-01	Missing guard / barrier	Machine operated without protective guard
UC-02	Defective equipment	Crane safety component was defective
UC-03	Poor housekeeping	Oil spill created a slip hazard
UC-04	Inadequate lighting	Work area had insufficient lighting
UC-05	Gas / hazardous atmosphere	Toxic gas present in work area
UC-06	Electrical hazard	Exposed electrical wiring
UC-07	Missing fall protection	Open edge without guardrail
UC-08	Inadequate exclusion zone	No physical separation around hazardous operation
UC-09	Unsafe pressure / energy condition	Equipment remained under hazardous pressure
UC-10	Inadequate ventilation	Confined area lacked adequate ventilation

3. UA + UC in the AI Safety System

Detection flow: Incident Report → NLP/Text Processing → Unsafe Act (UA) and/or Unsafe Condition (UC) → Hazard → Exposure → Barrier Analysis → SIF Potential (HIGH / MEDIUM / LOW).

4. Example

Incident: “A worker entered a confined space without gas testing and no injury occurred.”

Component	System Output
UA	Required entry procedure not followed
UC	Potential hazardous atmosphere
Exposure	YES
Critical Barrier	Atmospheric Gas Testing
Barrier Status	MISSING
SIF Potential	HIGH (subject to model/rule evaluation)

5. Key Principle for Problem Statement 26165

The system must not decide SIF potential from injury keywords alone. It should consider the combination of hazard, exposure, and critical barrier failure. Therefore, a report can have HIGH SIF potential even when “no injury occurred.”

6. Presentation Summary

Unsafe Act (UA): Human action that violates a safety procedure or increases hazard exposure.

Unsafe Condition (UC): Physical, environmental, equipment, or system condition that creates or increases hazard exposure.

AI decision logic: UA/UC → Hazard → Exposure → Barrier Failure → SIF Potential.